

A hybrid bio-inspired optimization algorithm for UAV-UGV cooperative path planning

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Abstract—This paper investigates a path planning approach for cooperative unmanned aerial vehicle (UAV) and unmanned ground vehicle (UGV) systems. Considering the position error corrections (PEC) of UAV, threats, and other complex constraints, a UAV-UGV cooperative path planning model is developed to minimize the total cost. Then, a Hybrid Ant-Pigeon Cooperative Optimization (HAPCO) algorithm is presented to address issues in existing bio-inspired optimization algorithms. A new pheromone update strategy and adaptive inertia weight are designed to optimise the performance. The performance of the Ant Colony Optimization (ACO) algorithm and Pigeon-inspired Optimization (PIO) algorithm. Moreover, the two algorithms are integrated to perform a full-cycle search. Simulation results reveal that the HAPCO algorithm outperforms ACO and PIO regarding path quality, cooperation cost, convergence, and computational efficiency, further demonstrated the validity and superiority of our algorithm.

Index Terms—path planning, unmanned autonomous systems, bio-inspired optimization, obstacle avoidance

I. INTRODUCTION

Artificial intelligence-enhanced unmanned autonomous systems are capable of performing tasks in complex environments. They are employed extensively in military, industrial, and civilian contexts [1]–[4]. In particular, the capacity of UAV and UGV to undertake tasks collaboratively has introduced novel tactical advantages to contemporary military operations. Effective collaboration between UAV and UGV not only enhances the efficiency of task execution but also improves operational flexibility and safety.

Path planning, as a key technology for the cooperative task execution of UAVs and UGVs, has recently emerged as a prominent area of research [5], [6]. Ropero et al. [7] proposed a path planning method that combines the Voronoi diagram and gravitational optimization for UAV-UGV cooperative exploration tasks. Zhang et al. designed a series of improved A-Star algorithms to achieve proactive threat avoidance for aircraft [8]–[10]. However, these methods are characterised by high computational complexity and limited flexibility when dealing with high-dimensional optimization problems. Inversely, bio-inspired optimization algorithms, due

to their adaptability and efficiency, have demonstrated strong performance in collaborative path planning, such as ant colony optimization (ACO) [11], particle swarm optimization (PSO) [12], pigeon-inspired optimization (PIO) [13], and genetic algorithm (GA) [14]. Duan et al. [15] proposed a dynamic discrete pigeon-inspired optimization algorithm that achieves multi-UAV cooperative task allocation and path planning. However, significant challenges remain when dealing with more complex constraints and threats.

To address this gap, this paper develops a UAV-UGV cooperative path planning model incorporating multiple complex threats and constraints. Considering the limitations of traditional bio-inspired optimization algorithms, we propose a hybrid bio-inspired optimization-based approach to achieve optimal, collision-free paths for UAVs and UGVs while effectively handling various constraints. Finally, we evaluate the performance of our model and algorithm and discuss their potential in practical applications. The contributions of this paper can be summarized as follows.

- 1) We establish a UAV-UGV cooperative path planning model under multiple constraints with predictive paths as variables, better reflecting real-world UAV-UGV scenarios.
- 2) To settle the cooperative path planning model and obtain optimal, collision-free path, we propose the HAPCO algorithm to improve global search capability, and effectively address the problem of getting trapped in local optima.
- 3) The effectiveness and optimality of the model and HAPCO algorithm are verified by designing simulations in complex scenarios. Compared to conventional bio-inspired algorithms, our method demonstrates faster convergence and meets the mission demand for autonomous systems.

II. PROBLEM FORMULATION AND SYSTEM MODEL

A. Problem Formulation

In military operations, UAVs perform wide-area reconnaissance, surveillance, and penetration missions, while UGVs carry out ground tasks like target search and material transport. Cooperative path planning between UAVs and UGVs is crucial for improving task execution efficiency and combat effectiveness. This section considers a cooperative system involving one UAV and multiple UGVs. A schematic diagram of the UAV-UGV cooperative path planning is depicted in Fig. 1.

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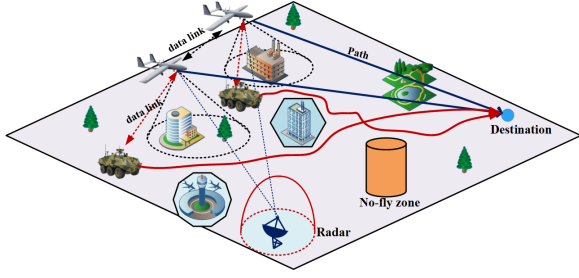


Fig. 1. A schematic diagram of cooperative path planning for the UAV-UGVs.

B. Kinematics

The kinematic model of the UAV is given by [16]

$$\begin{cases} \dot{x} = v \cos \theta \cos \psi \\ \dot{y} = v \cos \theta \sin \psi \\ \dot{z} = v \sin \theta \\ \dot{\psi} = (r \cos \phi + q \sin \phi) / \cos \theta \\ \dot{\phi} = p + \tan \theta (r \cos \phi + q \sin \phi) \\ \dot{\theta} = q \cos \phi - r \sin \phi \end{cases} \quad (1)$$

where (x, y, z) is the position of the UAV in the ground coordinate system. v is the velocity. ψ is the yaw angle. ϕ is the roll angle. θ is the pitch angle. r, p, q represent the yaw rate, roll rate, and pitch rate, respectively.

Similarly, the kinematic of UGV is given by

$$\begin{cases} \dot{x}_i = v_i \cos \theta_i \\ \dot{y}_i = v_i \sin \theta_i \\ \dot{\theta}_i = \frac{v_i}{L} \tan \gamma \end{cases} \quad (2)$$

where x_i, y_i is the position of the UGV, θ_i is the heading angle, L is the distance between the wheels. γ is the steering angle.

C. The Penetration Mission of UAV

The UAV needs to fly from the starting point to the endpoint to execute its mission. Due to inherent positioning errors, it must continuously correct these errors during the flight. Additionally, the UAV must avoid radar threats to ensure the successful completion of the mission. The objective function for the UAV performing the penetration mission is given by

$$U = \sum_{i=1}^n \sum_{j=1}^n x_{ij} \left(d_{ij} + p_f e_{ij} \right) + \sum_{k=1}^M \alpha_0 e^{-\frac{d_k}{R_k}} + \sum_{s=1}^Q \frac{\beta_0}{d_s + \epsilon} \quad (3)$$

where the first term is the path cost of the UAV, the second term denotes the cost of position error corrections (PEC), and the sum of the last two terms is the threat cost. n is the number of waypoints of the UAV. $x_{ij} \in \{0, 1\}$ is whether the UAV chooses to take the path from point i to point j . d_{ij} is the distance between i and j . $e_{ij} \in \{0, 1\}$ is whether the error correction for the path from point i to point j fails, p_f is the probability of failure. M is the number of radars. d_k is the distance from the UAV to the center of the k -th radar. R_k is

the detection radius of the k -th radar. Q is the number of no-fly zones. d_s is the distance from the UAV to the surface of the s -th no-fly zone. ϵ is a minimal positive number. α_0 and β_0 are threat coefficient of radars and no-fly zones, respectively.

D. The Search Mission of the UGV

Considering the path, threat, and position coordination costs with the UAV. The objective function for the UGV is as

$$S = \sum_{i=1}^h \left(D_i + \sum_{k=1}^M \alpha_1 e^{-\frac{d_{ik}}{R_k}} + \sum_{s=1}^Q \frac{\beta_1}{d_{is} + \epsilon} + \beta_2 \|\mathbf{x} - \mathbf{x}_i\|^2 \right) \quad (4)$$

where D_i is the path cost of the i -th UGV, h is the number of UGV. d_{ik} is the distance from the i -th UGV to the center of the k -th radar. The second and third terms are similar to those in Eq.(3). $\beta_2 \|\mathbf{x} - \mathbf{x}_i\|^2$ represents the coordination cost. d_{is} is the distance from the i -th UGV to the surface of the s -th no-go zone. $\mathbf{x}$ is the position vector of UAV, $\mathbf{x}_i$ is the position vector of the i -th UGV. α_1, β_1 , and β_2 are the threat coefficient of radar, no-go zone and collaborative cost, respectively.

E. UAV-UGV Cooperative Path Planning Model

According to Eqs. (1) to (4), the UAV-UGV cooperative path planning model is given by

$$J = \min (\omega_1 U + \omega_2 S) \quad (5)$$

Subject to

$$\begin{cases} \sum_{i=1}^n \sum_{j=1}^n x_{ij} \leq 1 \\ \sum_{i=1}^n \sum_{j=1}^n e_{ij} \leq \delta \\ \psi_{\min} \leq \psi \leq \psi_{\max} \\ \phi_{\min} \leq \phi \leq \phi_{\max} \\ \theta_{\min} \leq \theta \leq \theta_{\max} \\ d_k, d_{ik} \geq R_k, \quad d_s, d_{is} > 0 \\ \omega_1 + \omega_2 = 1 \end{cases} \quad (6)$$

where δ is the upper bound of the error. ω_1 and ω_2 are the cost weights for the UAV and UGV, respectively.

III. HYBRID ANT-PIGEON COOPERATIVE OPTIMIZATION ALGORITHM FOR PATH PLANNING

In this section, we propose a hybrid ant-pigeon cooperative optimization (HAPCO) algorithm to solve the UAV-UGV cooperative path planning model and obtain an optimal and collision-free path. Moreover, ACO and PIO is integrated to enhances the global search capability and efficiency, preventing it from becoming trapped in local optima.

A. The Analysis of ACO and PIO Algorithm

Ant colony optimization (ACO) algorithm is a probabilistic-based bio-inspired optimization algorithm [17]. The state transition probability on the map is given by

$$P_{ij}^k(t) = \frac{\tau_{ij}^\alpha(t) \eta_{ij}^\beta(t)}{\sum_{s \in C} \tau_{is}^\alpha(t) \eta_{is}^\beta(t)} \quad (7)$$

where $\mathcal{C}$ is the set of following nodes that the ant j can choose. $\tau_{ij}^\alpha(t)$ is the pheromone concentration for the ant moving from point i to point j . α is the pheromone influence factor. β is the heuristic influence factor. $\eta_{ij} = (D_{ij} + D_{jG})^{-1}$ is the heuristic value for moving from point i to point j . D_{ij} is the distance between point i and point j . D_{iG} is the distance from the point j to the destination G . The pheromone of ant is as

$$\tau_{ij}(t+1) = (1-\rho)\tau_{ij}(t) + \sum_{k=1}^{N_a} \Delta\tau_{ij}^k(t) \quad (8)$$

where ρ is the pheromone evaporation coefficient. N_a is the number of ants. $\Delta\tau_{ij}^k(t) = Q_a/L_k$ denotes the additional pheromone concentration deposited by ant k . Q_a is the pheromone intensity. L_k is the path length of ant k .

The pigeon-inspired optimization (PIO) algorithm is a bio-inspired algorithm that mimics the homing behavior of pigeons. In the two homing phases, pigeons rely on a map and compass operator in the early stage and a landmark operator in the later stage to approach the destination [18]. In the initial phase, the pigeon uses the map and compass operator for searching, and the number of iterations is N_c .

The velocity $\mathbf{V}_m(t)$ and position $\mathbf{X}_m(t)$ is as

$$\begin{cases} \mathbf{V}_m(t) = \mathbf{V}_m(t-1)e^{-Rt} + \text{rand}(\mathbf{X}_g - \mathbf{X}_m(t-1)) \\ \mathbf{X}_m(t) = \mathbf{X}_m(t-1) + \mathbf{V}_m(t) \end{cases} \quad (9)$$

where $R \in [0, 0.1]$ is the map and compass operator. $\mathbf{X}_g$ is the position of the best pigeon in the flock during the iteration cycle. $\text{rand}(\cdot) \in (0, 1)$ is a random number.

When the pigeons is close to the destination, their navigation relies on the landmark operator due to the influence of the surrounding environment. The state update strategy is as

$$\begin{cases} N_p(t) = \text{ceil}\left(\frac{N_p(t-1)}{2}\right) \\ \mathbf{X}_c(t) = \frac{\sum_{i=1}^{N_p(t)} \mathbf{X}_m(t) \text{Fit}(\mathbf{X}_m(t))}{N_p(t) \sum_{m=1}^{N_p(t)} \text{Fit}(\mathbf{X}_m(t))} \\ \mathbf{X}_m(t) = \mathbf{X}_m(t-1) + \text{rand}(\mathbf{X}_c - \mathbf{X}_m(t-1)) \end{cases} \quad (10)$$

where $\mathbf{X}_c(t)$ is the position of the central pigeon in the flock. $\text{Fit}(\mathbf{X}_m(t))$ is the fitness value of each pigeon. $N_p(t)$ is the number of pigeons. $\text{ceil}(\cdot)$ denotes the ceiling function.

B. Design Scheme of the HAPCO Algorithm

Despite the advantages of the ACO algorithm in local search and pheromone guidance, it exhibits weak global search capabilities and is prone to getting trapped in local optima. PIO algorithm excels in global search and rapid convergence but cannot find acceptable local searches under complex constraints [19]. To overcome the limitations, a HAPCO algorithm is developed to tackle global and local optimization challenges.

Firstly, in the PIO algorithm with a fixed number of pigeons N_p , the inertia weight $\omega(t) = e^{-Rt}$ in Eq. (9) implies that each pigeon is assigned the same weight, which fails to reflect

individual differences. Over time, the PIO algorithm may fall into the trap of local search. To further enhance the global search capability of the PIO algorithm in the initial phase, we designed an adaptive inertia weight that can dynamically adjust based on the fitness differences among different pigeons, thereby improving the speed at which the flock converges to the optimal solution. The adaptive inertia weight is defined as

$$\omega(t) = \begin{cases} e^{\frac{-\mathbf{V}_m(t-1)Rt}{2v_{\min}}}, & \text{if } \mathbf{V}_m(t-1) < v_{\min} \\ e^{\frac{-\text{Fit}(\mathbf{X}_m)Rt}{\text{Fit}(\mathbf{X}_g)}}, & \text{if } v_{\min} \leq \mathbf{V}_m(t-1) \leq v_{\max} \\ e^{\frac{-v_{\max}Rt}{\mathbf{V}_m(t-1)}}, & \text{otherwise} \end{cases} \quad (11)$$

Then, a new pheromone update strategy is introduced to improve the search capability of the ACO algorithm and enhance the convergence speed of the algorithm in the later stages. The increment of pheromone on the current path is as

$$\Delta\tau_{ij}^k(t) = \frac{L_{\text{best}} - L_k}{L_{\text{best}}} \quad (12)$$

where L_{best} is the length of the optimal path.

In the new pheromone update strategy, the pheromone concentration on the current optimal path can be dynamically adjusted based on the conditions of the paths searched by the ants. When a better path exists, a significant difference between L_k and L_{best} , which indicates that the proportion of ants traversing the current optimal path is tiny. Therefore, the intensity of the pheromone on the current optimal path can be dynamically increased to attract more ants by using Eq. (12).

As the iterations progress, the pheromone concentration on the current optimal path will continuously increase, causing more and more ants to tend to cluster along this path. At this point, the pheromone concentration will subsequently decrease. When the pheromone concentration reaches zero, the pheromone evaporation occurring solely on the current optimal path will weaken the pheromone, effectively preventing the ACO algorithm from getting trapped in local optima.

A negative pheromone concentration can reduce the accumulated pheromone concentration on the path and shorten the time required to search for the optimal path. This dynamic pheromone updating strategy allows the pheromone concentration to play a full role on the current optimal path, while preventing the algorithm from stagnating due to an excessive increase in its concentration.

Furthermore, we integrate the ACO algorithm and the PIO algorithm. Based on Eqs. (5) and (6), the HAPCO algorithm is applied to solve the path planning model. The implementation of the HAPCO algorithm is outlined as follows.

Step 1. Initialization: Input the starting point and destination of the UAV and UGV. Set the relevant parameters for the PIO and ACO algorithms.

Step 2. Global path search based on the PIO algorithm: In the initial phase, update the speed and position of the UAV and UGV according to Eqs. (9) and (11). Record the global optimal solutions $\mathbf{X}_g$ and update the global optimal cost S . Then, continue executing the PIO algorithm.

Step 3. **Algorithm switch:** Determine whether the current time t has reached the maximum iteration time N_c for the map and compass operators in the PIO algorithm. If it is, use the (x, y, z) and x_i, y_i as inputs to execute the ACO algorithm. Otherwise, return to Step 2.

Step 4. **Local path optimization based on the ACO algorithm:** Generate the number of ants based on the global path nodes obtained from the PIO. Initialize the position of each ant according to these nodes. Repeatedly execute the ACO algorithm to update the path nodes, i.e., $j = \arg \max_{s \in \mathcal{C}} P_{ij}^k(t)$. Then, update $\tau_{ij}(t+1)$ according to Eq. (12).

Step 5. **Path evaluation and coordination:** At the time t , update the path $\mathbf{P}_{ant} = \{\mathbf{P}_{uav}, \mathbf{P}_{ugv}\}$ and global cost S , and compare it with the current optimal path. Determine whether the cost of the path is optimal. If it is, record the current state and proceed to Step 4. Otherwise, ignore the current node and select other nodes from set $\mathcal{C}$ to continue searching. The algorithm will stop until the target is reached. Finally, output the optimal path $\mathbf{P}_{ant}^* = \{\mathbf{P}_{uav}^*, \mathbf{P}_{ugv}^*\}$ and the cost S^* .

IV. NUMERICAL RESULTS

In this section, to validate the effectiveness of the proposed model and the HAPCO algorithm, simulation experiments for UAV-UGV cooperative path planning are conducted in a complex scenario.

A. Experimental Setup

We consider a scenario involving a UAV and two UGVs within a $10000 \text{ m} \times 10000 \text{ m} \times 4000 \text{ m}$. The starting points and destination for both the UAV and UGVs are given, and some parameters are set as follows. $v = 80 \text{ km/h}$. $v_i = 40 \text{ km/h}$. $\{\psi, \theta_i\} \in [0, \pi/3]$. $\{\phi, \theta\} \in [0, \pi/2]$. $\alpha = 2$. $\beta = 1$. $\omega_1 = 0.6$. $\omega_2 = 0.4$. $N_a = 50$. $N_p = 50$. $N_c = 200$.

B. The Trajectory of UAV and UGVs

We conducted simulation experiments using the HAPCO, ACO algorithm [17], and PIO algorithms [18]. The UAV-UGV cooperative path planning results are illustrated in Fig. 2, where Figs. 2(a) to 2(c) display the 3D paths, and Figs. 2(d) to 2(f) present the paths from a top view.

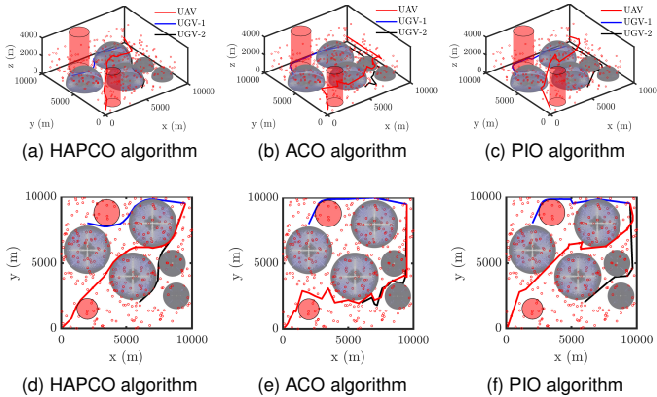


Fig. 2. Path planning results. (a)~(c): 3D paths. (d)~(f): Top view.

In Fig. 2, the gray areas are radar threats, while the red areas indicate no-fly zones for the UAV and UGV. The small red circles denote the error correction points of the UAV. The path planning results suggest that all three algorithms provide collision-free paths for both the UAV and UGV, ultimately enabling them to reach their target points. The UAV utilizes these error correction points to complete its trajectory planning while rapidly ensuring positioning accuracy, and both unmanned vehicles effectively avoid environmental threats. Notably, compared to the ACO and PIO algorithms, the HAPCO algorithm yields higher-quality paths.

C. Performance Evaluation and Algorithm Comparison

We show the performance of the proposed HAPCO algorithm in terms of path optimality, computational efficiency, UAV-UGV collaboration, and handling of complex constraints by selecting several critical metrics from the cooperative planning model, as shown in Tab. I. Additionally, we provide the distances between the UAV and UGVs in Fig. 3.

TABLE I
THE PERFORMANCE METRICS OF DIFFERENT ALGORITHMS.

Algorithm	Path cost (km)	Number of PEC	Cooperative cost	Run time (s)
HAPCO	31.137	6	5.75×10^4	69.07
ACO	40.235	14	9.4×10^4	132.93
PIO	35.281	19	7.22×10^4	97.58

Tab. I shows that the HAPCO algorithm has significant advantages regarding total path cost, the number of position error corrections, cooperative cost between UAV and UGV, and run time. Specifically, compared to the ACO and PIO algorithms, the HAPCO algorithm reduces the path cost by 29.2% and 13.3%, respectively. The number of error corrections decreases by 133.3% and 216.7%. The cooperative cost is reduced by 63.5% and 25.6%, and the run time is decreased by 92.5% and 41.3%, respectively. These results validate the effectiveness and optimality of the HAPCO algorithm.

Fig. 3 depicts that the HAPCO algorithm can achieve UAV-UGV cooperative path planning in less time. Additionally, the distance curves between the UAV and UGV obtained using the HAPCO algorithm exhibit significantly lower amplitude and frequency of variation than those obtained by the ACO and PIO algorithms. This further indicates a strong level of cooperation between the UAV and UGVs, which validates the superiority of the proposed HAPCO algorithm.

D. Convergence Analysis

We evaluated the convergence performance of different algorithms to validate the optimality of the solutions in UAV and UGV cooperative path planning. This includes the HAPCO, ACO, and PIO algorithms. The variation of the objective function J versus iteration time is depicted in Fig. 4.

Fig. 4 shows that the HAPCO algorithm achieves the minimum value for the objective function of cooperative path planning, confirming the optimality of the algorithm. During the

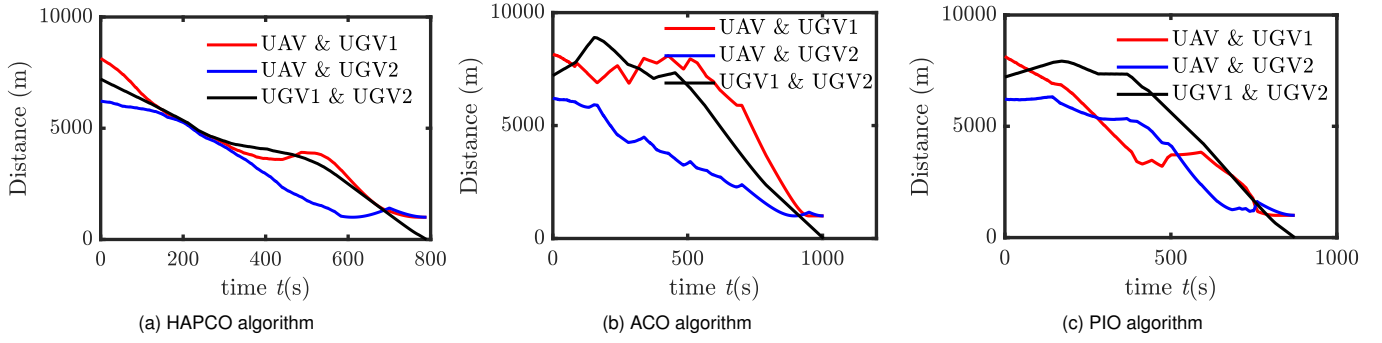


Fig. 3. Distance between UAV and UGV versus time.

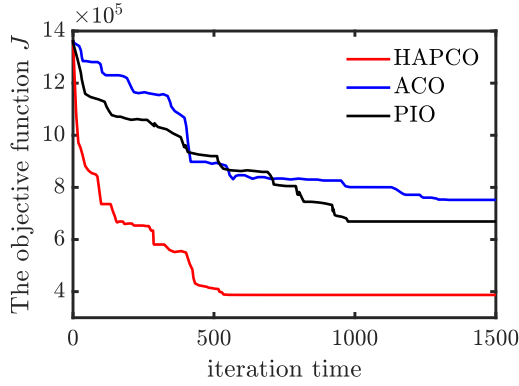


Fig. 4. The convergence curves of different algorithms.

optimization process, the convergence speed of the HAPCO algorithm is faster than the other two algorithms, indicating that the design approach of integrating and improving the search strategies of the ACO and PIO algorithms is feasible and effective. Additionally, while the ACO algorithm performs better during the mid-search phase, the PIO algorithm generally exhibits superior global search capability and convergence speed compared to the ACO algorithm.

V. CONCLUSION

This paper presents a Hybrid Ant-Pigeon Cooperative Optimization (HAPCO) algorithm to address the cooperative path planning of UAVs and UGV. Our algorithm enhances global search capabilities and convergence speed. Compared to the ACO and PIO algorithms, simulation results show that the proposed HAPCO algorithm reduces path costs by 29.2% and 13.3%, decreases the number of positioning error corrections by 133.3% and 216.7%, enhances computational efficiency by 92.5% and 41.3%, and lowers cooperative costs by 63.5% and 25.6%. These findings validate the effectiveness and superiority of our model and algorithm.

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